OEAS 705/805 Advanced Environmental Data Science

Fall 2026

Week 1

Dr. Jessie Turner, jturners@odu.edu

Class times: OCNPS 320, T/Th, 11:00 - 12:15

Office hours: After class T/Th,

And: OCNPS 411, M, 1:00 - 3:00

Survey for the Start of the Semester

- In your email.
- Anonymous.
- Let me know if you do not have the link.

Advanced Environmental Data Science Survey

Fall 2026
Instructor: Jessie Turner (jturners@odu.edu)

This survey is designed to give me an understanding of students' level of coding experience upon the start of the semester so I can teach the group accordingly. This survey is completely anonymous. We will repeat the survey at the end of the semester to monitor our growth.

* Indicates required question

Please enter your birth year followed by the last two digits of your phone number (e.g., 197095). This will serve as your anonymous unique identifier to track growth from the start to the end of the semester. *

Your answer

AI Policy Recap

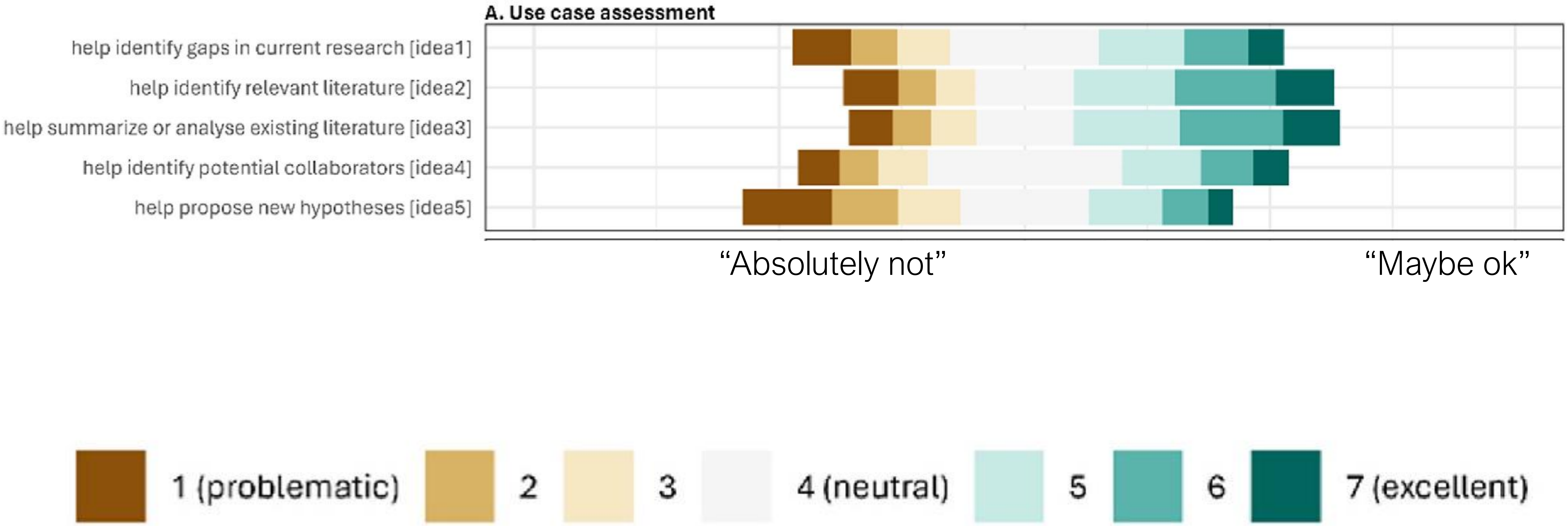
- Use AI for coding.
- Credit the tool.
- Provide the full dialogue with the tool.

Example:

- Google Gemini, log in using MIDAS ID
- Check privacy settings
- Prompt, get answers
- Select all
- Copy and paste into a text file

AI Policy Recap

- Use AI for coding.
- AND understand pros and cons:



2000+ Researchers in Denmark responded to this survey
Andersen et al. 2025, Technology in Society

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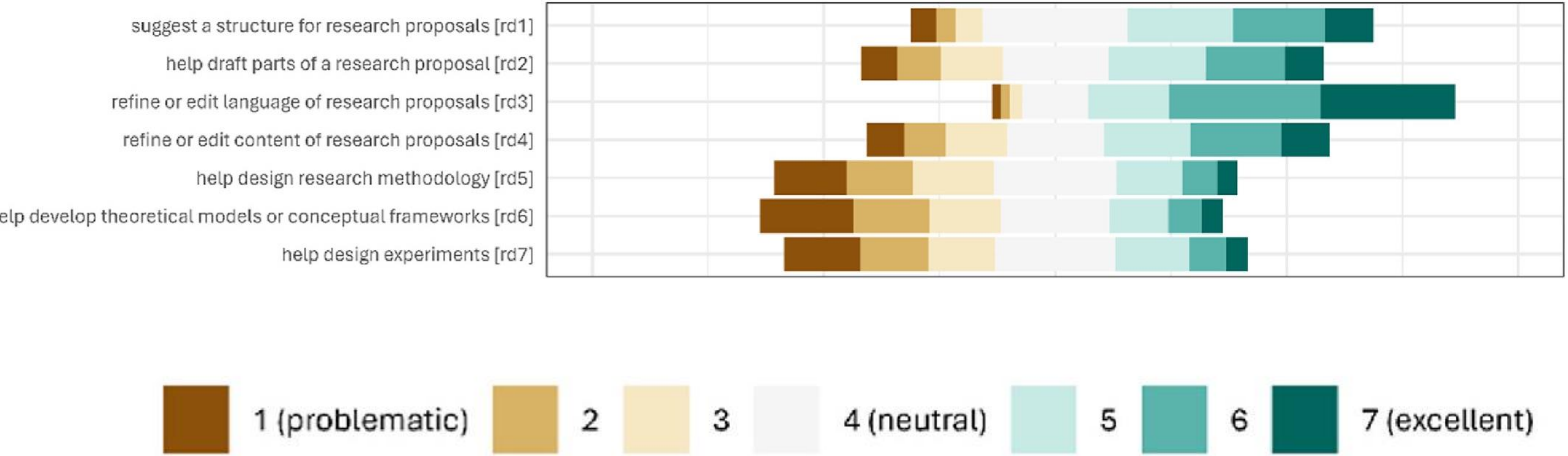

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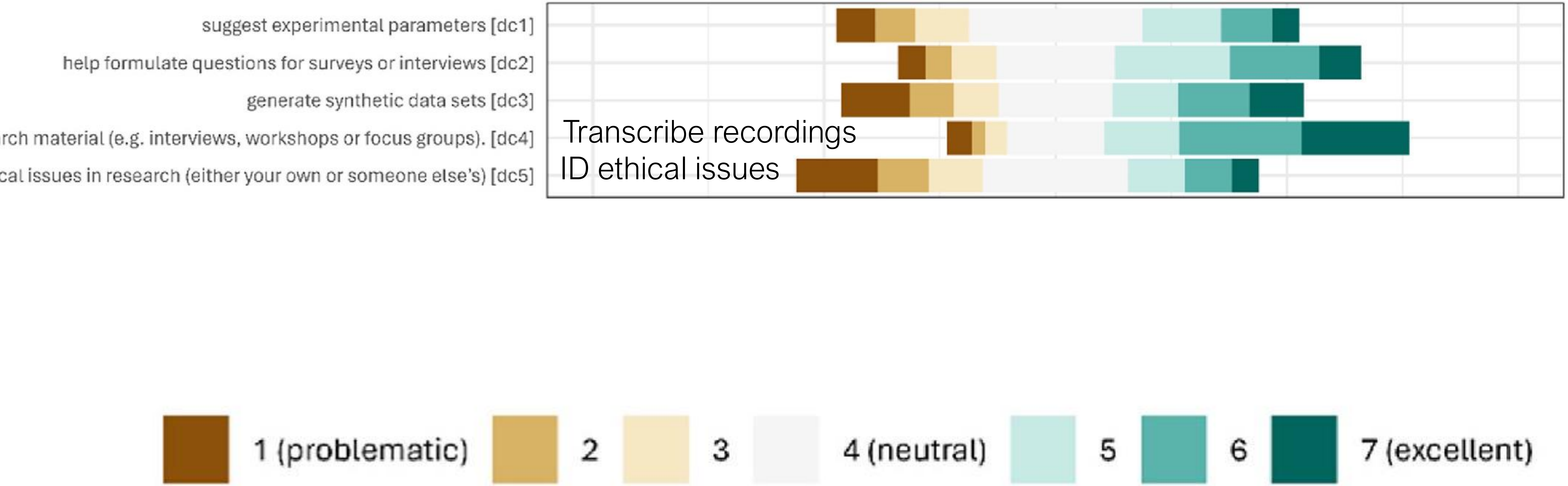

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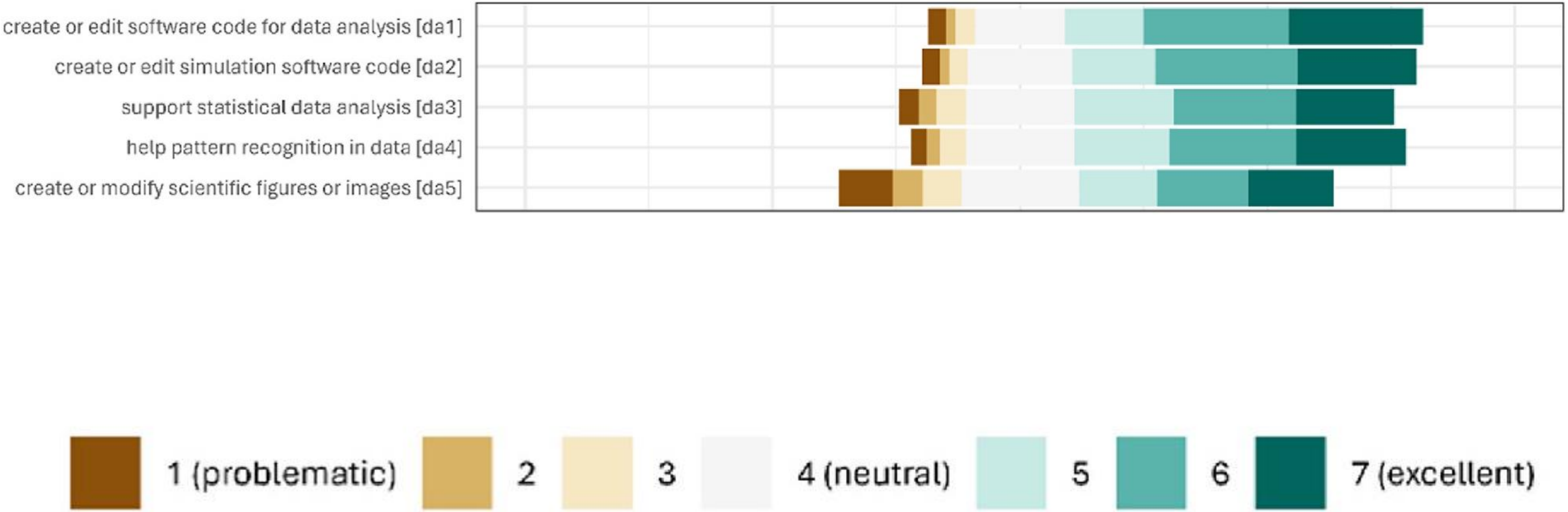

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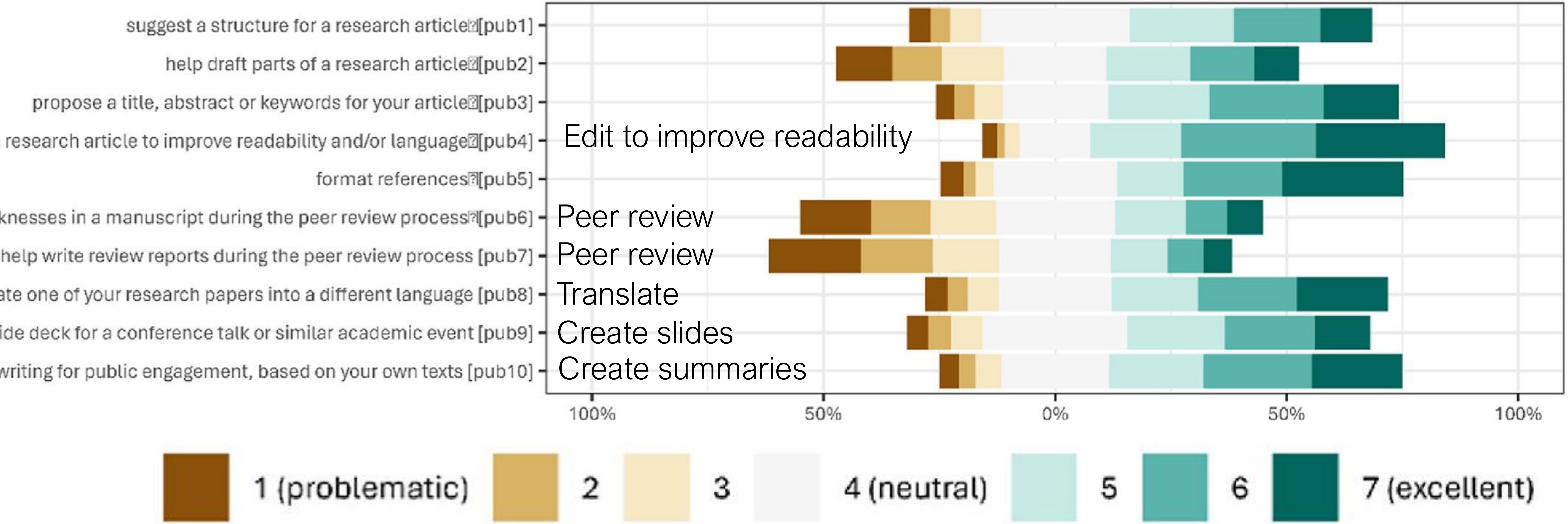

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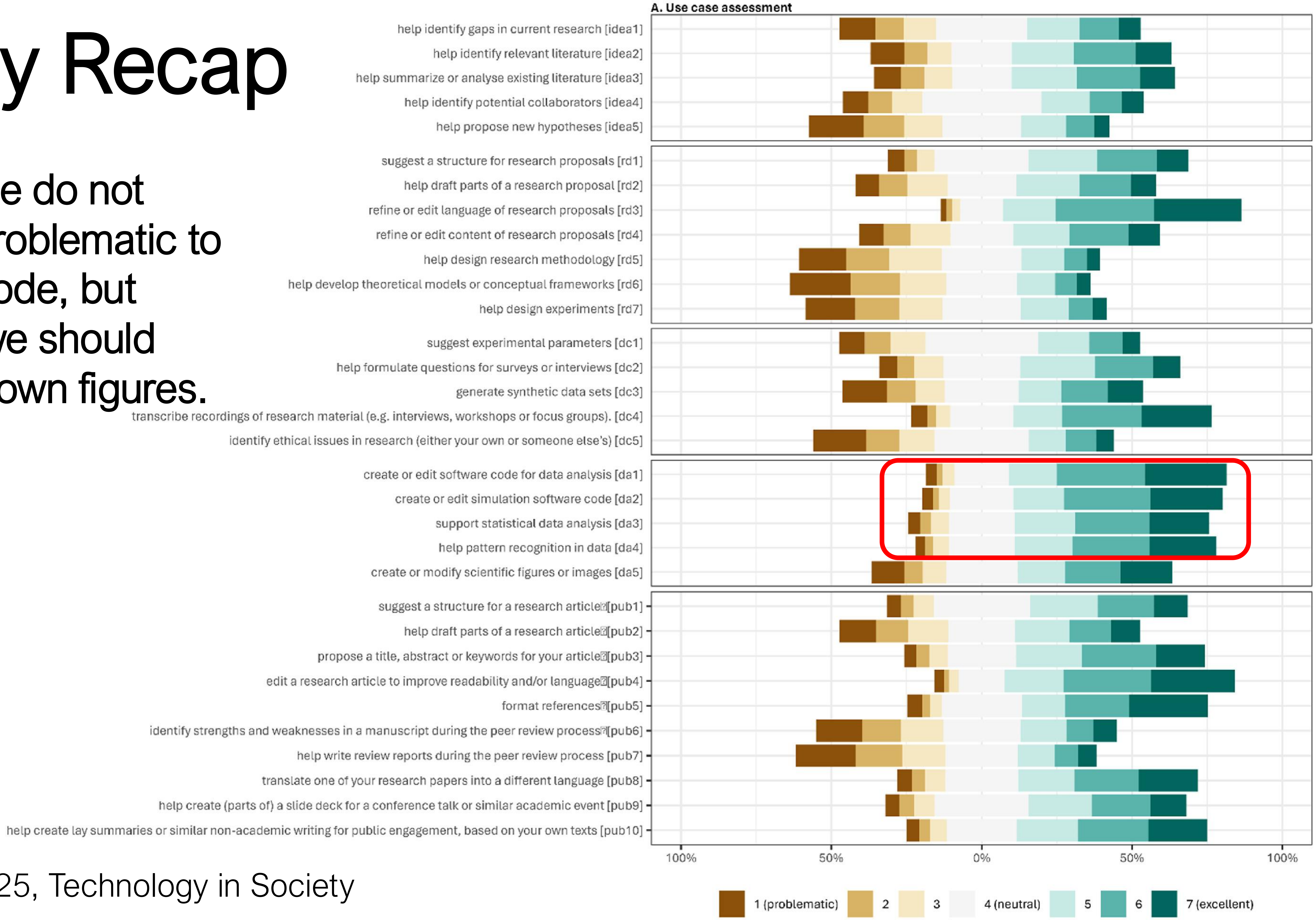
AI Policy Recap

Most people do not think it is problematic to use AI to code, but think that we should create our own figures.



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FAIR data principles review

F = FINDABLE

A = ACCESSIBLE

I = INTEROPERABLE R

= REUSABLE

FAIR data principles review

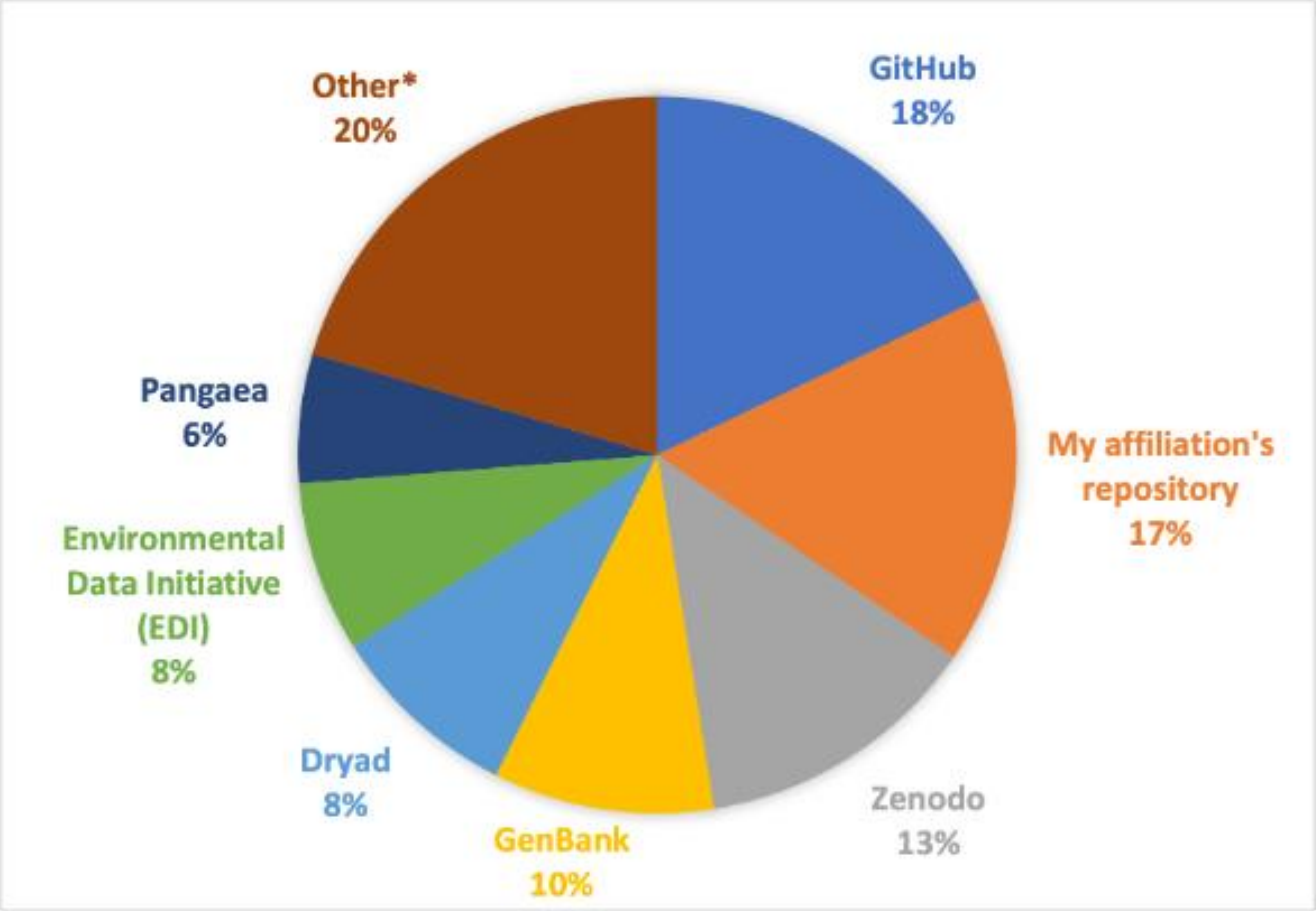
Where are aquatic science researchers publishing their datasets?

- Institutional data repository? (ODU Digital Commons)
- Government agency data repository?
- Independent data repository?

Survey conducted by past RCE Fellows Erin Peck and Rita Franco-Santos in 2024 on ASLO members' awareness of and opinions about Open Access publishing:

- Of N=239 responses, 54 unique data repositories.
- Of the 54 unique data repositories, 19 of them were run by government agencies (U.S. and other nations)

FAIR data principles review



Survey data courtesy of Rita Franco-Santos and Erin Peck.
*Other: includes government agency data repositories

FAIR data principles review

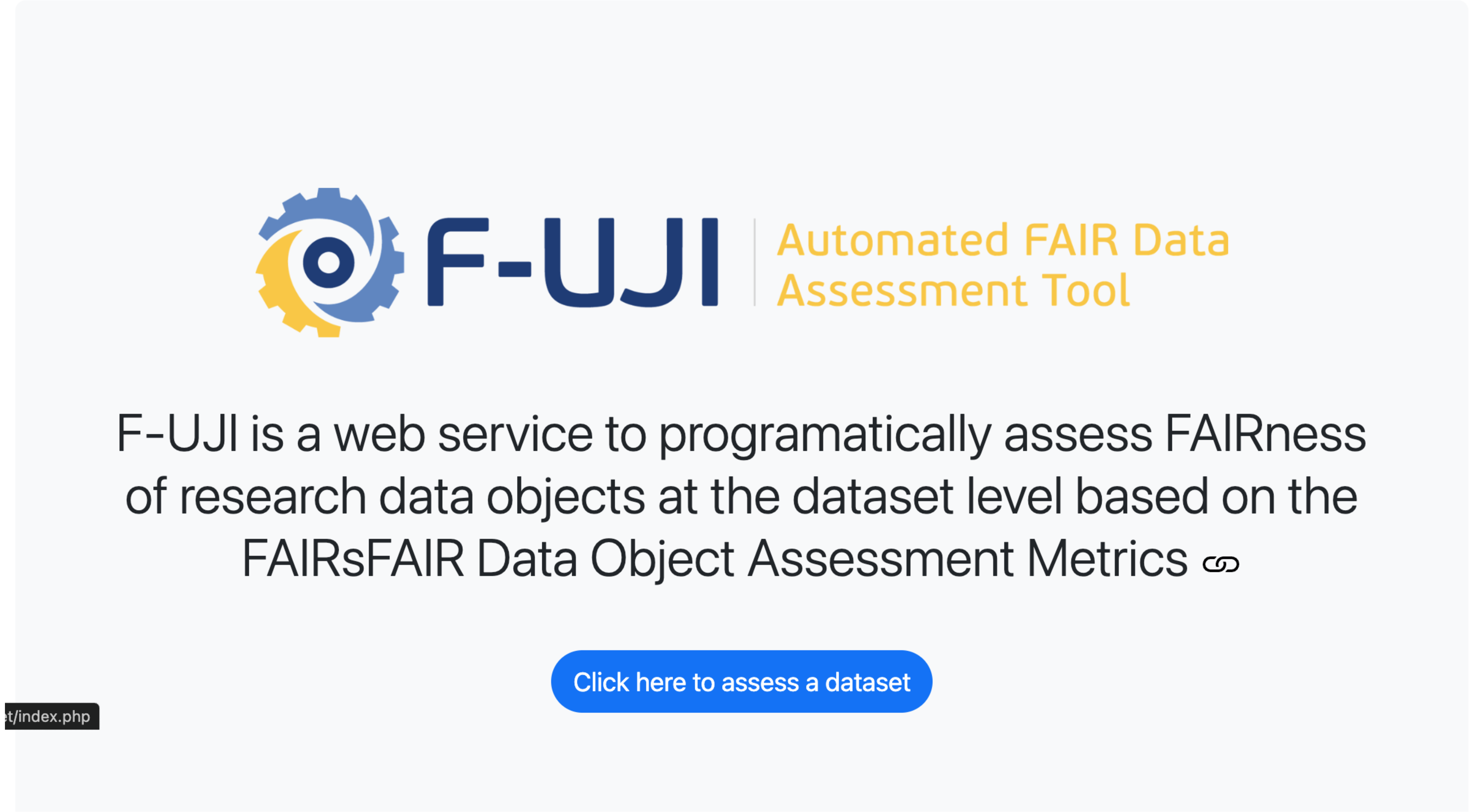
<u>DIFFERENT THINGS:</u>

- DOI – Digital Object Identifier
- Dataset – All the data associated with a project
- Data repository – hosts open-access datasets with DOIs
- GitHub repository – open, but not permanent, no DOI
- Data paper – published in a journal, describes a dataset

FAIR data principles review

Check the “FAIRness” of a dataset:

https://www.f-uji.net/



Course Housekeeping

Tools: - laptop
- git
- GitHub
- conda
- python
- Interface (Virtual Studio Code / Jupyter Notebook)
- Text editor (Sublime Text or BBEdit or other)

We are starting here

FIRST: Command Line basics

Course Housekeeping

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We are starting here

GitHub page for the class > notes > git_exercise

Before Tuesday

HW 1 Due Monday by midnight

Find it on the GitHub page for the class

Upload the link to your own GitHub page to Canvas